The Jane Street Enforcement Shock: Retail Behavior in India's Index Options Market

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Abstract

This paper examines retail trader responses to the July 2025 SEBI interim order against Jane Street, which banned the quantitative trading firm from Indian markets for alleged index manipulation involving *Rs.*4,843 crores ($576.6 million USD) in impounded gains. Using an event study methodology with 120-day baseline estimation, I estimate a regression model controlling for market returns, liquidity, volatility, and expiry dynamics to identify abnormal changes in retail participation and risk exposure following the enforcement action. The findings reveal that retail participation increased substantially with a cumulative abnormal return of 26.89 percentage points by day seven, while retail put exposure decreased significantly with a cumulative abnormal return of -13.41 percentage points. These results challenge existing theoretical frameworks developed primarily in Western institutional contexts. Rather than uniformly retreating from regulatory uncertainty or blindly seeking volatility, retail traders exhibited selective responses - expanding market presence while simultaneously reducing specific risk exposures. This study contributes novel insights into retail behavior in emerging market derivatives and has implications for regulatory design in retail-dominated markets.

1 Introduction

The Indian stock market is currently the fourth largest in the world with a market capitalization of $5.13 trillion USD (World Bank, 2024), following the United States, China, and Japan. Remarkably, India reached this significant position within 33 years of establishing the National Stock Exchange in 1992.

1.1 Historical and Regulatory Foundations (1991–2010)

The nation's economic liberalization in 1991 gave rise to this rapid development. Key reforms were deployed in direct response to a severe balance of payments crisis in 1991, which depleted India's foreign exchange reserves to a critical level. Decades of heavy government oversight and regulation were discarded for initiatives to integrate the economy with the global market and increase privatization and foreign investment (Panagariya, 2016; International Monetary Fund, 2004).

There are currently two predominant exchanges in India: the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE). The former is one of the oldest stock exchanges in the world, established in 1875. However, today, the NSE comprises 94% of the current market share and 87% share in the equity options market in India (CNBC-TV18, 2024). The NSE's superior technological capabilities, transparent electronic trading system, higher volumes and liquidity, and a wider range of derivatives products allows it to dominate the market (Marketfeed, n.d.).

The Securities and Exchange Board of India (SEBI) was constituted as a non-statutory body in 1988. Through the provisions of the Securities and Exchange Board of India Act of 1992, the institution became a statutory body with powers to develop and enforce regulations within the law.

The introduction of derivatives in India further transformed the economy. For most of the post-independence period, the government had banned forward trading. In 1996,

the L.C. Gupta Committee associated with SEBI, in response to the NSE's proposal to list exchange-traded derivatives, recommended a phased introduction of derivative products (Sarkar, 2006).

As a result, the Securities Contracts (Regulation) Act of 1956 was amended to classify derivatives as "securities," thereby extending the existing regulatory framework to derivatives and dismantling the ban on these products.

Between 2000 and 2001, India introduced its first index futures, followed by index options and single-stock futures. The early 2000s saw a phased introduction of stock options, more index products, longer expiry contracts, and increased institutional participation and proprietary trading.

1.2 The Rise of Retail and Market Transformation (2010–2024)

Between 2010 and 2014, low-cost brokers such as Upstox (then RKSV Securities) and Zerodha launched retail services that greatly lowered the barrier to entry for trading. Digitization, Aadhaar-based eKYC, and the proliferation of affordable internet access via Jio fundamentally transformed market accessibility and participation (The Borgen Project, 2020).

In 2020, COVID-19 massively changed the ecosystem. Retail dominance in the derivatives market increased exponentially. Comparable to the U.S. Robinhood boom, India saw a parallel surge. Retail activity in options—especially weekly index options—skyrocketed because they appeared cheap with potential for high returns. At the macro level, interest rates hit all-time lows of 5–6%, inflation rose, and households accumulated excess savings due to lockdown-driven reductions in consumption and travel.

Regulatory and structural enablers included the introduction of weekly expiries, SEBI's simplified onboarding, and UPI (Unified Payments Interface) innovations enabling instant funding. Social media further democratized financial literacy—Instagram and YouTube "finfluencers" and WhatsApp options-selling groups encouraged individuals to participate in the markets, often without highlighting risk (Zerodha, 2024).

SEBI studies reveal that 93% of individual traders incurred losses in equity futures and options between 2022 and 2024, with aggregate losses exceeding $Rs.$1.8 Lakh Crores or $21.69 billion USD (Securities and Exchange Board of India [SEBI], 2024).

1.3 Foreign and Quant Participation in the Modern Market (2010–Present)

Structural and regulatory developments also accelerated foreign investment. Foreign Institutional Investors (FIIs) are companies established outside India that invest in Indian financial markets by registering with SEBI (Mishra, 2005).

According to the Central Depository Services (India) Limited (CDSL), FII investment in India has increased by approximately 758.3% since 1999. After the 1992 reforms, FII participation deepened with the launch of the National Securities Depository Limited (NSDL), facilitating paperless settlements and accelerating registrations.

In 2013, SEBI merged all FIIs, sub-accounts, and qualified foreign investors (QFIs) into a new investor class known as "Foreign Portfolio Investors" (FPIs), streamlining participation in index derivatives, currency derivatives, and interest rate futures (SEBI, 2013).

In 2016, the NSE launched weekly Bank Nifty and Nifty index options, further increasing demand for short-dated index derivatives from FPIs.

After COVID-19, high-frequency trading firms and global quantitative players expanded heavily into India. By then, SEBI had created one of the most arbitrage-friendly major markets in the world—deep index futures liquidity, weekly expiries, high retail mispricing, transparent regulations, and maturing electronic infrastructure.

IMC began trading in India in the late 2000s but formally opened a Mumbai office in 2021. Citadel entered the market with an office in Gurgaon in 2022. In 2024, Optiver began trading in India. Firms such as AQR, Two Sigma, and Millennium expanded operations as well. Quantitative trading giant Jane Street Group began operating in India in December 2020. The firm has an estimated AUM of $140 billion USD.

1.4 Understanding Index Options in India

Derivatives are financial instruments whose value derives from an underlying asset, such as a stock index, individual stock, commodity, or currency. In India, the most actively traded derivatives are index options. These financial products grant the holder the right, but not the obligation, to buy or sell a basket of stocks represented by an index at a predetermined price (referred to as the strike price) on or before a specific expiration date. Index options in India are primarily written on Nifty 50, which tracks the 50 largest companies on the NSE, and Bank Nifty, which specifically tracks banking sector stocks. These contracts are settled in cash based on the difference between the strike price and the index closing value at expiration.

There are two types of index options in this market: call options and put options. A call option gives the buyer the right to profit from upward index movements. This effectively functions as a bullish bet. On the other hand, a put option provides the right to profit from downward movements, serving a bearish speculation or portfolio hedge. The seller or writer of these options collects premium income upfront but assumes the obligation to pay out if the option finishes in-the-money at expiration. In-the-money options have intrinsic value and would be profitable if exercised immediately, at-the-money options have strike prices near the current index level, and out-of-the-money (OTM) options have no intrinsic value and would be worthless if exercised today. OTM options are particularly popular among retail traders in India because they require minimal upfront capital, often costing just a few hundred rupees per contract.

For example, a far out-of-the-money put option on Bank Nifty might cost $Rs.50$ but could theoretically return $Rs.10{,}000$ or more if a market crash drives the index below the strike price before expiration. This "lottery-like" payoff structure and weekly expiries that create frequent opportunities for speculation, have made OTM index options the dominant vehicle for retail participation in India's derivatives market. With low entry costs and high leverage, derivatives trading volumes far exceed those in equity cash markets. In fact, according

to the National Institute of Securities Markets, India's average daily traded value in the equity-derivatives segment (EDS) in FY 2024–25 was *Rs.*2.63 lakh crore ($31.3 billion USD), compared to *Rs.*1.2 lakh crore ($14.3 billion USD) in the cash (equity) market.

1.5 Jane Street-SEBI Shock: Research Question and Hypothesis

In July 2025, SEBI issued an interim order banning Jane Street from trading in Indian markets for alleged index manipulation (SEBI, 2025). The order outlines claims of intraday index manipulation, marking-the-close behavior, taking losses in cash/futures to gain on larger options books, and continuing the behavior despite NSE's February 2025 warning. SEBI ordered the impoundment of over *Rs.*4,843.57 crores ($576.6 million USD) in alleged illegal gains.

This paper investigates how the July 2025 interim SEBI order against Jane Street altered the retail-dominated index options market beyond intra-day effects. I examine the question: *To what extent did the July 2025 SEBI–Jane Street–driven shock affect retail participation and retail risk exposure in India's index options market?* This question is motivated by the unprecedented scale of alleged manipulation in BANKNIFTY index options, the regulatory uncertainty that followed, and the uniquely retail-heavy composition of India's short-dated derivatives markets.

The original hypothesis was that retail participation and retail risk exposure would decrease following the July 2025 Jane Street shock. This hypothesis was grounded in evidence that retail investors exhibit loss aversion, reduce leverage after volatility shocks, and behave more conservatively when institutional activity is uncertain. However, the empirical results of this study contradict this prediction on retail participation.

The postulated thesis suggests that the Jane Street-SEBI shock increased retail participation and decreased risk exposure, reflecting India's distinctive combination of low-friction trading infrastructure, retail-dominant index options liquidity, and strong volatility-seeking behavior among small traders.

2 Literature Review

The existing literature on retail participation, derivatives microstructure, and market responses to institutional shocks can be organized into three competing schools of thought: (1) the Structural Stability School, which argues that retail behavior in India's index options market is largely invariant to institutional disruptions; (2) the Retail Retreat School, which predicts that retail traders reduce participation following volatility shocks; and (3) the Volatility-Seeking School, which suggests that retail participation may rise in response to heightened uncertainty and large payoffs.

2.1 Structural Stability School

The Structural Stability School argues that retail participation does not meaningfully change after institutional disturbances. In the Indian context, this is substantiated by the retail-dominated composition of the index options market, richness of liquidity, and persistent trading routines.

Chordia, Roll, and Subrahmanyam (2001) suggest that liquidity and trading activity tend to be jointly driven by broad market conditions rather than isolated institutional events, implying that micro-level institutional shocks have limited power to shift aggregate participation patterns. Similarly, activity in emerging-market derivatives often reflects structural demand and macro exposure rather than event-driven changes, suggesting stability amidst market stressors.

Evidence from Indian derivatives markets specifically is consistent with this structural view. Jain, Mishra, and Tantri (2017) show that small investors participate consistently even under regulatory intervention. This is largely because derivatives offer low-cost, cash-efficient exposure that retail traders continue to demand despite external shocks. Their policy experiment repeatedly finds persistent retail engagement, supporting the argument that India's weekly index options ecosystem is structurally sticky. Barber and Odean (2000)

demonstrate that individual investors exhibit habitual trading patterns that persist across market conditions, suggesting that routine-based behavior can dominate over event-driven responses. In the Indian context, the infrastructure developed between 2010 and 2020, including zero-commission brokers, UPI-based instant funding, and mobile trading platforms, created deeply embedded participation routines. The weekly expiry cycle itself became a structural feature of retail trading behavior, with participation concentrated around regular settlement windows regardless of institutional developments.

Furthermore, Hasbrouck and Saar (2013) find that in highly liquid markets with diverse participant bases, the impact of individual institutional players is often absorbed quickly, with minimal lasting effects on aggregate trading patterns. India's index options market, with its extraordinary retail dominance and deep liquidity pools, may exhibit similar shock-absorption characteristics. The sheer volume of retail participation creates a form of structural inertia that dampens the influence of any single institutional actor, even one as large as Jane Street.

Thus, in adherance with the Structural Stability School, it can be predicted that the July 2025 Jane Street driven shock would not significantly alter retail participation or risk exposure, because retail behavior is anchored in stable routines and market structure rather than institutional or regulatory shocks.

2.2 Retail Retreat School

The Retail Retreat School argues that retail participation decreases following major shocks, driven by heightened perceived risk, margin pressure, and loss aversion. This view draws heavily from developments in behavioral finance. Pan and Poteshman (2006) suggest that retail investors adjust options positions in response to order-flow imbalances and uncertainty, reducing aggressive trading as perceived risk increases due to loss aversion and disproportionate sensitivity to negative events.

The retreat hypothesis is grounded in prospect theory and reference-point dependence.

Kahneman and Tversky (1979) establish that losses appear larger than gains for most decision-makers, creating asymmetric responses to positive and negative information. In derivatives markets, this asymmetry is amplified by leverage, margin requirements, and the complexity of options payoffs. Shefrin and Statman (1985) extend this framework to show that investors exhibit temperament changes and risk-reduction following losses or increased uncertainty. Dhingra, Gandhi, and Bulsara (2012) find that, in the Indian markets, retail traders become more conservative following volatility shocks, exhibiting reduced risk-taking and diminished participation consistent with broader findings on loss-aversion dynamics.

Additionally, De Silva, So, and Smith (2025) show that retail trading intensity declines when institutional flows or volatility shocks introduce uncertainty into derivatives markets. This is because retail traders lack the sophistication to arbitrage complex informational signals. The informational disadvantage of retail participants becomes more prominent during regulatory events or when the strategic behavior of sophisticated or institutional players is unclear.

The retreat school also emphasizes the role of margin constraints and liquidity pressures. Brunnermeier and Pedersen (2009) demonstrate that funding liquidity and market liquidity are mutually reinforcing, creating spirals during stress periods. For retail traders operating with high leverage in weekly index options, a sudden increase in volatility can trigger margin calls, forced liquidations, and cascading position reductions. SEBI's interim order and the immediate impact of the Jane Street-shock resulted in such reactions for retail traders.

In addition, research on trust and market participation suggests that regulatory events involving allegations of manipulation can erode confidence. Guiso, Sapienza, and Zingales (2008) find that trust is a critical determinant of stock market participation, and that scandals or enforcement actions can have persistent negative effects on retail engagement. The Jane Street case, with its allegations of systematic manipulation and impoundment of over $575 million USD, could reasonably be expected to trigger trust-based withdrawal from the affected markets.

Together, these findings predict that, according to the Retail Retreat School, the July 2025 shock should have lowered both retail participation and retail put exposure because retail traders tend to retreat from markets during periods of uncertainty, manipulation risk, and regulatory ambiguity.

2.3 Volatility-Seeking School

Finally, the Volatility-Seeking School argues that retail participation increases in the presence of volatility, attention shocks, and large, lottery-like payoffs. These conditions are often created by institutional crises or market-wide shocks.

Barber and Odean (2007) demonstrate that retail investors buy attention-grabbing securities such those demanding headlines or with unusual activity. Thus, an enforcement action against a major market-maker like Jane Street likely raises visibility and draws retail interest into the market. Barber and Odean (2008) extend this framework to show that individual investors are net buyers of stocks following extreme one-day returns, news events, and high trading volume. These scenarios tend to be prevalent in the aftermath of major regulatory interventions.

The attention mechanism is particularly powerful in the modern Indian context, where financial information spreads rapidly through social media, WhatsApp groups, and YouTube channels. DellaVigna and Pollet (2008) show that retail trading responds disproportionately to salient and easy-to-process information rather than complex fundamental analysis. Dorn and Sengmueller (2009) suggest that entertainment-motivated trading increases with market uncertainty, and that retail traders exhibit preference reversals that favor risky, exciting positions during periods of heightened attention. The Jane Street case, with its dramatic enforcement action and significant media coverage, created this type of attention shock that drives retail engagement.

In addition, Bogousslavsky and Muravyev (2024) show that sharp volatility spikes around market-close events generate option-return asymmetries and short-term opportunities that

disproportionately attract speculative traders. This suggests that a shock increasing settlement volatility could increase small-trader activity. Kumar (2009) provides substantiating evidence that retail investors exhibit strong "lottery preferences" where they gravitate toward securities with positively skewed return distributions and "jackpot-like" payoff structures. These characteristics are prevelant in India's weekly index options, particularly in out-of-the-money contracts.

These volatility-driven entries and exits create a never ending loop of irrational buying and selling. Barberis and Xiong (2009) suggest that investors who have experienced losses often increase risk-taking in an attempt to recover prior wealth levels. This behavior is particularly prominent in options markets where leverage can amplify potential recovery, but also add to losses.

The Indian market's unique structural features may further reinforce volatility-seeking behavior. The dominance of weekly expiries creates continuous opportunities for speculative positioning. Furthermore, the retail-heavy market structure reduces the stigma or informational disadvantage associated with small-trader participation. Unlike markets where retail traders are clearly at a disadvantage relative to institutions, India's options market has a more level playing field in terms of access, execution costs, and market impact. This can potentially encourage continued or increased participation even during uncertainty.

Thus, this school predicts higher retail participation and greater risk-taking following the July 2025 shock because retail traders seek to capitalize on volatility and market visibility amplified by Jane Street's trading ban.

2.4 Synthesis and Empirical Predictions

These three schools generate divergent predictions about retail behavior following the Jane Street shock. The Structural Stability School predicts no change in participation or risk exposure, the Retail Retreat School predicts decreases in both, and the Volatility-Seeking School predicts increases in both. The analysis in this study is designed to identify which

of these competing hypotheses are reflective of actual retail behavior. This is accomplished by examining actual changes in retail participation rates, put-to-call ratios, and trading intensity across the National Stock Exchange's index-options following the July 2025 SEBI order.

It is important to note that the India-specific context adds complexity to these predictions. The market's retail dominance, zero-commission structure, instant funding capabilities, and social-media-driven participation culture create conditions that are unlike those studied in most Western derivatives markets. Furthermore, India accounts for about 80% of the world's index and stock options contracts, playing a significant role in the global derivatives market. The SEBI enforcement action itself was unprecedented in scale, involving allegations of systematic manipulation worth over $575 million USD and targeting one of the world's largest quantitative trading firms. These features suggest that the Indian case may provide novel insights into retail behavior that extend beyond the existing theoretical frameworks.

3 Methodology

3.1 Case Selection & Justification

The Jane Street–Bank Nifty episode constitutes a critical case for examining FII influence on retail participation. Critical case selection, as defined in comparative methodology, identifies instances where theoretical predictions should hold most strongly. If phenomena cannot be observed in critical cases, theoretical frameworks require revision (Gerring, 2007).

SEBI's investigation into Jane Street produced unprecedented documentation of institutional trading patterns during index-options expiry periods. Regulatory reports detail specific trading strategies, position sizes, timing of trades, and profit magnitudes - information typically unavailable to researchers studying institutional behavior. Thus, this episode provides a rare window into the micro-level processes through which institutional trading

might affect retail participation.

One potential limitation of this case-based approach is that India's derivatives market exhibits structural characteristics quite different from other markets: incredibly high derivatives-to-cash turnover ratios (300:1), dominant retail participation in options trading, and zero-day-to-expiry contract availability. These features limit generalizability to other market contexts. Thus, findings from this research illuminate dynamics within India's unique market structure while suggesting hypotheses for testing in other emerging market derivatives contexts.

3.2 Data Sources

The primary market data are obtained from the National Stock Exchange (NSE) of India and Bloomberg Terminal. The NSE participant-wise open interest reports categorize OI into four categories: Foreign Institutional Investors (FII/FPI), Domestic Institutional Investors (DII), retail clients, and proprietary traders.

Expiry day identification for the isExpiry_t binary indicator is obtained directly from NSE's calendar API, which provides authoritative dates for futures and options expiries across all index products. This approach ensures precise alignment with actual settlement schedules and avoids potential misclassification that could arise from manual calendar construction or rule-based approximations.

All the remaining data are sourced from Bloomberg Terminal. Daily index values (open, high, low, close) are collected for all five major NSE indices employed in the turnover-weighted return calculation: BANKNIFTY, FINNIFTY, MIDCPNIFTY, NIFTY, and NIFTYNXT50. Trading volumes across each index were extracted to create an aggregate liquidity variable. Daily market turnover for each index guided the development of an aggregate market return variable. India VIX data was also utilized to measure market-wide implied volatility.

A limiting factor in the data structure is that there is no granularity on individual firms'

trading behaviors (such as Jane Street). Thus, FPI behavior is analyzed through a broader categorical lens rather than distinguishing between strategies across individual firms. Similarly, retail behavior data are aggregated at the market level, with no information available on individual trader positions, transaction histories, volume distributions, or prior experience levels. This aggregation is consistent with privacy protections in Indian securities regulation but constrains the ability to examine heterogeneity within the retail trader population. Furthermore, there is no Bank Nifty specific open interest data, requiring the scope of the analysis to extend beyond the shock parameters itself and into the broader index-options market.

Regulatory and qualitative data supplement the quantitative analysis by adding deeper context as to the strategies employed by Jane Street and the market reception. SEBI's interim order dated July 2025 and subsequent regulatory filings provide detailed information about the alleged manipulation activities, the scale of impounded gains, and the specific trading patterns that triggered enforcement action. Financial press coverage from sources including The Financial Times, Economic Times, and Business Standard documents market participant reactions, institutional responses, and broader industry implications of the Jane Street case. Jane Street's official response documents submitted during the regulatory proceedings offer the firm's perspective on trading strategies and compliance interpretations. These qualitative materials provide essential context for interpreting the quantitative results and understanding the mechanisms through which the regulatory shock may have affected retail behavior.

Data quality and coverage are generally high for both NSE and Bloomberg datasets, reflecting India's advanced electronic trading infrastructure and comprehensive regulatory reporting requirements. The participant-wise open interest data from NSE are reported daily with standardized classifications, ensuring consistency across the sample period. Bloomberg Terminal provides institutional-grade data with rigorous quality controls and consistent reporting standards across all index products. NSE's calendar API delivers programmatically

accessible, machine-readable expiry schedules that eliminate manual data entry errors.

The January 15, 2025 to July 2, 2025 pre-event window contains 120 complete trading days with no missing observations for the primary variables. Post-event data extend through the remainder of 2025, providing sufficient observations to test for sustained changes in retail participation and risk exposure patterns following the SEBI enforcement action.

3.3 Event Study and Model Specification

This study employs an event study approach to examine retail behavior in India's index options market following the July 2025 SEBI enforcement action against Jane Street. The first stage establishes baseline behaviors and relationships between retail participation, risk exposure, and key market variables during the pre-event period. The second stage tests whether these relationships changed significantly after the shock.

The analysis focuses on two dependent variables that capture distinct and complementary dimensions of retail trader behavior. RetailShare_t measures retail participation as the percentage of total open interest or trading volume attributable to retail traders.

$$\text{RetailShare}_t = \frac{\text{Retail OI or Volume}_t}{\text{Total OI or Volume}_t} \tag{1}$$

RetailPutShare_t measures retail risk exposure as the ratio of retail-held put positions (both long and short) to total retail open interest across all index options. This captures how retail traders express directional views and hedge market risk.

$$\text{RetailPutShare}_t = \frac{\text{Retail Put OI}_t^{(\text{long + short})}}{\text{Total Retail OI}_t} \tag{2}$$

The baseline regressions are as follows:

$$\text{RetailShare}_t = \alpha + \beta_1 \cdot \text{WtdMarketReturn}_t + \beta_2 \cdot \text{IndiaVIX}_t + \beta_3 \cdot \text{isExpiry}_t + \beta_4 \cdot \text{Liquidity}_t + \varepsilon_t \tag{3}$$

$$\text{RetailPutShare}_t = \alpha + \beta_1 \cdot \text{WtdMarketReturn}_t + \beta_2 \cdot \text{IndiaVIX}_t + \beta_3 \cdot \text{isExpiry}_t + \beta_4 \cdot \text{Liquidity}_t + \varepsilon_t \tag{4}$$

Where WtdMarketReturn_t represents turnover-weighted returns across NSE indices, IndiaVIX_t captures market volatility, isExpiry_t is a binary indicator for futures and options expiry days, and Liquidity_t represents aggregate options trading volume across major indices.

3.4 Construction of Independent Variables

The model incorporates several refinements to standard market microstructure variables. These definitions are adapted to the distinctive features of India's derivatives market and based on data availability.

The market return variable is calculated using a turnover-weighted aggregation approach rather than relying on a single index return. This is because open interest data as provided by the National Stock Exchange is aggregated across all index-options. The weighting approach adapts the logic of activity-based aggregation from Chordia, Roll & Subrahmanyam (2001) and Bekaert, Harvey & Lundblad (2007), who show that weighting financial variables by trading intensity, such as turnover or volume, provides a more accurate measure of market-wide conditions. The turnover-weighted return is constructed as follows:

$$w_i = \frac{\sum_j \text{Turnover}_j}{\text{Turnover}_i} \tag{5}$$

$$\text{WtdMarketReturn}t = \sum_i w_i \cdot \ln(Ri, t) \tag{6}$$

This incorporates returns from five major NSE indices: BANKNIFTY, FINNIFTY, MID-

CPNIFTY, NIFTY, and NIFTYNXT50. Weighting the returns by turnover ensures that more actively traded indices exert a proportionally greater influence on the constructed market factor. NSE indices account for over 99% of all index derivatives traded in India while BSE's share is under 1% and effectively zero for most index-option products. Thus, restricting the model to NSE indices ensures economic relevance and is ideal when considering data availability constraints.

The liquidity variable is constructed by aggregating options trading volume across the major index-options: NIFTY, BANKNIFTY, FINNIFTY, and NIFTYNXT50. Although NIFTYNXT50 contributes a smaller share than the dominant three (NIFTY, BANKNIFTY, FINNIFTY), including it ensures that the liquidity measure aligns with the indices represented in retail open interest for which data are significant and available. This approach is consistent with derivatives-market microstructure research, including Boguslavsky & Muravyev (2024), who demonstrate that option liquidity affects participation dynamics, price efficiency, and the relative activity of institutional versus retail traders. In markets where sophisticated liquidity providers (such as Jane Street) typically supply depth, liquidity shocks or shifts disproportionately affect retail participation. Thus, using aggregate options volume across all actively traded NSE index options provides a theoretically grounded and robust liquidity proxy.

IndiaVIX, the NSE's implied volatility index, is a critical control variable. Its inclusion as an independent variable is consistent with volatility-based models of trading behavior, where volatility shocks meaningfully shift both institutional strategies and retail investor activity. Kumar and Lee (2006) show that volatility measures capture distinct dimensions of market uncertainty that influence participation decisions. Garleanu, Pedersen, and Poteshman (2009) demonstrate that implied volatility indices contain forward-looking information about trader expectations and risk preferences.

The model also incorporates a binary expiry-day indicator (isExpiry_t). Expiry days have material influence on trading intensity, order flow, and settlement-related behavior.

3.5 Sample Period and Event Window Construction

The selection of an appropriate event window presents methodological challenges given India's dynamic regulatory environment. MacKinlay (1997) provides theoretical guidance for event studies, recommending a 120-day pre-event estimation period to model normal market conditions and establish baseline parameter estimates. However, the Indian regulatory landscape during 2024-2025 was characterized by frequent policy announcements, implementation delays, and timeline extensions that complicate the identification of a clean pre-event period.

On February 4, 2025, SEBI issued a circular on safer participation of retail investors in algorithmic trading. This regulatory change introduced new compliance standards with an initial implementation timeline, which was subsequently extended on September 30, 2025. This pattern of announcement, standard formulation, delayed implementation, and timeline extension represents a recurring regulatory cycle that SEBI has employed extensively, particularly over the past two years.

The high frequency of regulatory changes creates challenges for isolating the specific effect of the Jane Street enforcement action. The estimated effects of this study may partially reflect cumulative regulatory uncertainty occurring simultaneously in Indian markets.

Given these considerations, the final pre-event window spans January 15, 2025 to July 2, 2025, comprising 120 trading days. This period terminates immediately before the July 2025 Jane Street enforcement action, ensuring a clean break between pre-event and post-event periods. It also encompasses a period of relative regulatory stability compared to the dense cluster of policy changes that occurred later in 2025.

4 Results

4.1 Retail Participation (RetailShare$_t$)

The retail participation model achieves substantial explanatory power with an R^2 of 0.353, indicating that approximately 36% of the daily variation in retail share is captured by market returns, volatility, expiry effects, and liquidity conditions. After correcting for autocorrelation and heteroskedasticity using Newey-West standard errors, the regression reveals strong effects from volatility and liquidity variables.

Table 1: Baseline Regression Results: Retail Participation (RetailShare)

	Coefficient	Std. Error	t-statistic	p-value
Constant	59.8295	1.1847	50.4970	0.0000***
WtdMarketReturn	-12.7434	10.8421	-1.1754	0.2416
IndiaVIX	-0.5869	0.0849	-6.9111	0.0000***
isExpiry	0.5438	0.4446	1.2232	0.2232
Liquidity	-2.64e-07	7.95e-08	-3.3208	0.0011***
Observations	120			
R-squared	0.353			
Adjusted R-squared	0.331			
F-statistic	15.71			
Prob (F-statistic)	1.76e-10			

Note: Standard errors are Newey-West HAC standard errors.
*** p¡0.01, ** p¡0.05, * p¡0.1

IndiaVIX exhibits a strong negative association with retail participation (p ¡ 0.001). This suggests that retail traders systematically reduce their market share as volatility increases. This finding provides empirical support for the Retail Retreat School's prediction that small traders withdraw from derivatives markets during periods of heightened uncertainty. Therefore, it can be concluded that a one-standard-deviation increase in IndiaVIX corresponds to a measurable reduction in retail share relative to institutional participants.

Aggregate options trading volume (Liquidity$_t$) also displays a negative relationship with retail participation ($p = 0.001$). This suggests that institutional traders increase their relative presence during high-volume periods as compared to retail investors. Surges in market-wide

liquidity are disproportionately driven by institutional activity rather than retail expansion. When trading intensity increases across index-options, retail traders maintain absolute participation but lose market share to more active institutional players.

Neither turnover-weighted market returns nor the expiry-day indicator significantly influences retail participation in this baseline specification. This indicates that, during normal market conditions, short-term market returns do not shift retail share patterns. Similarly, the insignificant expiry coefficient suggests that while expiry days concentrate trading activity, they do not systematically shift the market's participation composition

4.2 Retail Risk Exposure (RetailPutShare$_t$)

The retail put share model demonstrates even stronger explanatory power with an R^2 of 0.441, indicating that market conditions, volatility, expiry effects, and liquidity jointly explain approximately 44% of the variation in retail put usage. Using heteroskedasticity and autocorrelation consistent (HAC) robust standard errors, the regression reveals that all four independent variables significantly influence how retail traders allocate positions between puts and calls.

Table 2: Baseline Regression Results: Retail Put Exposure (RetailPutShare)

	Coefficient	Std. Error	t-statistic	p-value
Constant	0.4828	0.0125	38.5728	0.0000***
WtdMarketReturn	0.2526	0.1004	2.5160	0.0132**
IndiaVIX	-0.0014	0.0009	-1.5556	0.1222
isExpiry	0.0105	0.0042	2.5000	0.0137**
Liquidity	5.77e-09	8.45e-10	6.8284	0.0000***
Observations	120			
R-squared	0.441			
Adjusted R-squared	0.422			
F-statistic	22.75			
Prob (F-statistic)	3.54e-14			

Note: Standard errors are HAC-robust (Newey-West).
*** p¡0.01, ** p¡0.05, * p¡0.1

Turnover-weighted market returns exhibit a positive and statistically significant relation-

ship with retail put share. This suggests that retail traders increase put exposure during bullish market periods. The positive coefficient captures a combination of hedging demand and premium-collection strategies that increase retail put positions relative to calls. Regarding the former, retail traders may purchase out-of-the-money puts as portfolio insurance when the market rallies, seeking to "lock-in" gains or protect against reversals. Secondly, rising markets may encourage speculative put selling among retail traders attempting to collect premiums from perceived overpriced downside protection.

IndiaVIX demonstrates a negative relationship with retail put share, indicating that higher volatility reduces the proportion of retail positions allocated to puts. This pattern contradicts simple hedging-based explanations, which would predict increased put demand during volatile periods. Instead, the negative volatility effect is consistent with retail traders avoiding expensive options when implied volatility is elevated. The cost of put protection rises sharply with IndiaVIX, potentially pricing retail traders out of bearish positions. This corrobarates the price-sensitivity of retail investors as they tend to reduce options activity when premiums become largely expensive.

The expiry-day indicator has a strong positive effect on retail put share. This suggests that retail put usage increases significantly on settlement days. Thus, speculative out-of-the-money put strategies are concentrated around weekly expiries. Retail traders in India's market frequently purchase short-dated, far out-of-the-money puts seeking lottery-like payoffs from sharp market declines near settlement. The concentration of these activities on expiry days creates a structural increase in retail put share, regardless of broader market conditions.

Market-wide liquidity exhibits a strong positive relationship with retail put share, indicating that retail risk-taking intensifies during high-volume periods. When aggregate options trading volume increases, retail traders allocate a larger proportion of their positions to puts, suggesting heightened speculative or hedging activity. This pattern may reflect retail traders' ability to access tighter bid-ask spreads and improved execution quality during liquid market

conditions. This reduces the transaction costs associated with put strategies.

Overall, the retail put share model captures sophisticated patterns in retail risk exposure that vary systematically with market conditions, volatility, settlement timing, and liquidity environments. The strong explanatory power suggests that retail options positioning is far from random, instead responding predictably to economic incentives and structural market features.

4.3 Event Study Results

The event study analysis examines abnormal changes in both retail participation and risk exposure following the July 3, 2025 SEBI enforcement order against Jane Street. Using the baseline regression parameters estimated over the pre-event window, abnormal returns are calculated:

$$AR_t = \text{Observed}_t - \text{Predicted}_t$$

Cumulative abnormal returns (CAR) track the persistent deviation from expected behavior over windows of [0,+1], [0,+3], [0,+5], and [0,+10] trading days.

$$CAR_{\tau_1,\tau_2} = \sum_{t=\tau_1}^{\tau_2} AR_t$$

4.3.1 Retail Participation Response

The event study reveals a clear, persistent increase in retail participation following the SEBI order. On July 3, 2025, the first trading day after the enforcement action, retail share reached 60.48%, representing a positive abnormal return of 2.65 percentage points relative to the model prediction. This initial shock intensified on July 4, with retail share climbing to 62.96% and abnormal participation of 5.13 percentage points. The cumulative abnormal return by day two reached 7.78 percentage points, indicating a substantial and sustained

shift in market composition.

Throughout the subsequent trading days, retail participation remained persistently elevated above model expectations. By July 7, the third trading day, CAR reached 12.85 percentage points. The pattern continued through the post-event window, with CAR reaching 23.91 percentage points by July 10 and 26.89 percentage points by July 11. The steady accumulation of positive abnormal returns demonstrates that the increase in retail participation was not a temporary reaction but rather a sustained structural shift lasting at over a week after the regulatory shock.

Table 3: Event Study Results: Cumulative Abnormal Retail Share

Date	Retail Share (%)	Abnormal Return	CAR	Trading Day
2025-07-03	60.475	2.646	2.646	+1
2025-07-04	62.961	5.131	7.777	+2
2025-07-07	62.906	5.076	12.853	+3
2025-07-08	62.606	4.776	17.629	+4
2025-07-09	61.648	3.818	21.448	+5
2025-07-10	60.295	2.465	23.913	+6
2025-07-11	60.802	2.973	26.886	+7

Note: Abnormal returns calculated as actual minus predicted values from baseline model. CAR = Cumulative Abnormal Return from July 3, 2025.

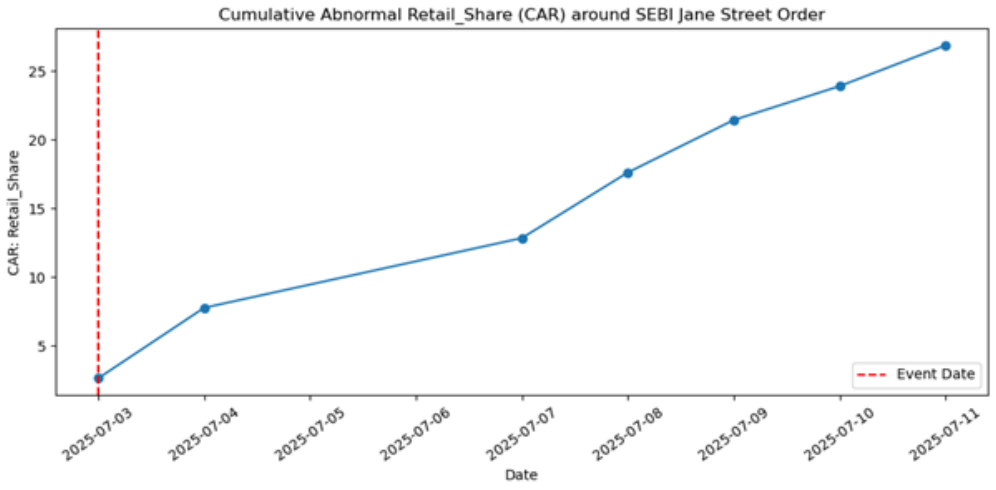


Figure 1: Cumulative Abnormal Retail Share (July 3 - July 11).

This pattern provides strong evidence to support the Volatility-Seeking School. Retail

traders not only maintained their presence despite the enforcement action but actively expanded their market share. They seemingly aimed to exploit the temporary reduction in institutional activity following Jane Street's forced exit. The rising CAR suggests that retail traders filled the participation gap created by the withdrawal or scale-down of institutional liquidity providers.

4.3.2 Retail Risk Exposure Response

The retail put share event study reveals an entirely different pattern. Retail traders systematically reduced their put exposure following the SEBI order. On July 3, retail put share registered a small negative abnormal return of 0.51 percentage points, indicating slightly lower put usage than predicted. However, this decline began to rapidly accelerate. On July 4, abnormal put share fell 1.58 percentage points and CAR reached -2.09 percentage points.

The reduction in relative put exposure intensified through the first week. Although a brief volatility-driven uptick occurred on July 8 (with positive abnormal put share of 1.26 percentage points), this was a temporary fluctuation. By July 9, CAR resumed its downward trajectory, reaching -5.10 percentage points. The decline continued sharply through the remainder of the event-window, with CAR falling to -13.41 percentage points by July 11.

Table 4: Event Study Results: Cumulative Abnormal Retail Put Share

Date	Retail Put Share	Abnormal Return	CAR	Trading Day
2025-07-03	0.4777	-0.0051	-0.0051	+1
2025-07-04	0.4670	-0.0158	-0.0209	+2
2025-07-07	0.4689	-0.0139	-0.0348	+3
2025-07-08	0.4954	0.0126	-0.0222	+4
2025-07-09	0.4540	-0.0288	-0.0510	+5
2025-07-10	0.4691	-0.0137	-0.0647	+6
2025-07-11	0.4135	-0.0694	-0.1341	+7

Note: Abnormal returns calculated as actual minus predicted values from baseline model. CAR = Cumulative Abnormal Return from July 3, 2025.

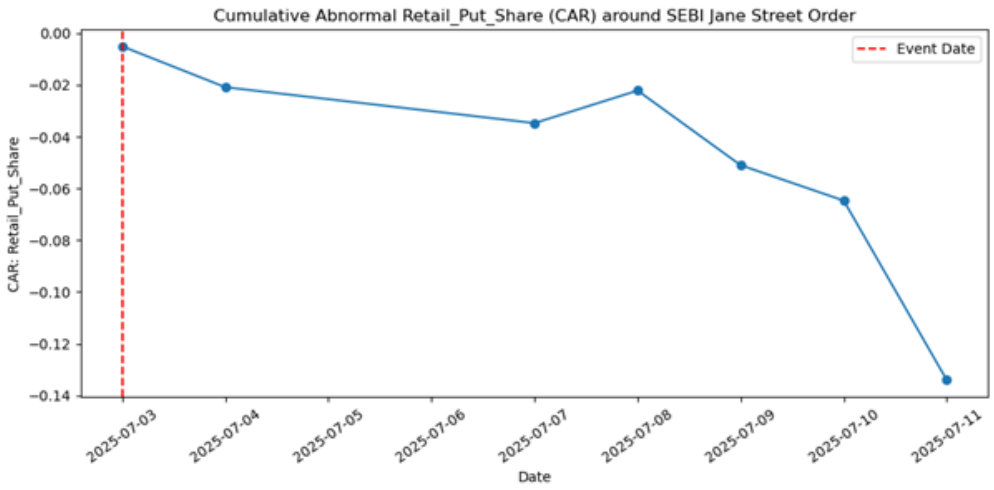


Figure 2: Cumulative Abnormal Put Share (July 3 - July 11).

This sustained reduction in retail put share suggests that, post-Jane Street shock, retail traders shifted their positioning away from bearish or speculative put strategies. The negative CAR trend indicates that retail traders became relatively more exposed to call options or reduced their hedging activity - though they maintained higher absolute participation levels.

5 Analysis and Discussion

5.1 Results Interpretation

The divergent patterns in retail participation and risk exposure provide nuanced evidence regarding the three competing theoretical frameworks. The results reject the strong form of the Retail Retreat School as retail traders did not withdraw from the market following the enforcement shock. However, the risk exposure results reveal a more subtle retreat mechanism. Retail traders maintained and even increased their market presence, but they systematically reduced exposure to riskier put positions that might have exposed them to losses if market conditions continued to deteriorate.

The sustained increase in retail participation supports elements of the Structural Stability School and Volatility-Seeking School. The market's retail-dominated structure proved

resilient to institutional disruption, and the magnitude of the participation increase suggests an active response to the shock. Retail traders seem to have perceived the Jane Street exit as an opportunity for speculation or as a catalyst in reducing competition from sophisticated institutional players.

The reduction in retail put share adds complexity to this interpretation. Rather than increasing risk-taking to exploit volatility opportunities, retail traders adopted a more conservative positioning strategy. Retail traders may have reduced hedging activity if Jane Street's exit implied reduced manipulation. If so, they viewed the market as safer and requiring less downside protection. An alternative explanation postulates that the enforcement action may have increased uncertainty about settlement prices and option payoffs, making directional call strategies more attractive than put positions.

The combination of increased participation and reduced put exposure suggests that the July 2025 SEBI order fundamentally altered retail trading behavior along multiple dimensions. Retail traders did not simply maintain existing patterns nor did they uniformly retreat or increase all forms of risk-taking. Instead, they exhibited multidimensional responses by expanding market presence while simultaneously reducing exposure to specific risk factors associated with bearish or speculative put strategies.

These findings contribute novel insights into retail behavior in derivatives markets during regulatory shocks. The results suggest that retail traders in India's market are neither purely passive participants nor uniformly unsophisticated actors who flee from uncertainty. Rather, they demonstrate selective responses to institutional disruptions, adjusting participation and risk exposures in an attempt to reflect changing market conditions and perceived opportunities. Though these endeavors predominantly result in monetary loss, such behavior reflects a persistent willingness among retail traders to engage with complex derivatives markets even when the odds are structurally unfavorable.

Broadly, however, the methodology cannot definitively establish causal mechanisms. While the event study identifies significant abnormal behavior following the SEBI enforce-

ment action, multiple channels could produce the observed patterns. Retail traders might respond to Jane Street's absence, to regulatory uncertainty, to media coverage of manipulation allegations, to changed market microstructure, or to coincident but unrelated factors. Disentangling these mechanisms would require additional data on trader beliefs, information sources, and decision processes.

5.2 Indian Market Perspective

In response to the Jane Street-SEBI regulatory shock, retail traders did not flee uncertainty, nor did they uniformly increase risk-taking. Instead, they exhibited selective responses, expanding participation while reducing exposure to specific risk factors. This finding suggests that existing theoretical frameworks, largely developed in Western market contexts, require modification when applied to retail-dominated emerging market derivatives.

The persistent increase in retail participation following Jane Street's trading ban indicates that institutional liquidity provision, while often beneficial, may also crowd out retail activity during normal market conditions. This suggests that, considering India's participation distribution, there may be latent demand that remains suppressed when dominant institutional players operate at scale. In fact, according to SEBI reports, Jane Street's trading activity on January 17, 2024 (a single trading day), comprised 25% of the market-wide traded value in the BANKNIFTY constituent stocks and futures. This mechanism has receives limited attention in existing literature, which typically assumes institutional participation enhances market quality without considering these potential substitution effects.

Furthermore, it is vital to characterize retail trading behavior through the lens of news coverage and social media - amplifying the dissemination of information beyond the pure issuance of the SEBI Interim Order. India's information ecosystem is vast and diverse. Major outlets including The Financial Times, Economic Times, and Business Standard emphasized the unprecedented scale of alleged manipulation (over $Rs.$4,843 crores in impounded gains) and the dramatic nature of marking-the-close strategies.

However, social media narratives often diverged from these factual reports, framing the event as evidence that "the big players were caught cheating" or that "retail investors now have a level playing field." WhatsApp trading groups and YouTube channels may have interpreted Jane Street's exit as creating profit opportunities previously monopolized by sophisticated institutions, encouraging increased participation through narratives of empowerment and revenge against manipulative "foreign money." This social amplification likely contributed to the attention-driven participation surge documented in the event study, as retail traders responded not to the technical details of the SEBI order but to simplified, emotionally resonant interpretations disseminated through trusted peer networks. The rapid behavioral response observed within days of the enforcement action suggests that perhaps, a more ardent driver in retail-deicision making was social media-mediated information diffusion, rather than careful reading of the 105 page SEBI order.

5.3 Policy Implications

These findings carry several implications for financial regulators, particularly SEBI and other emerging market authorities concerned with retail investor protection. The sustained increase in retail participation following the Jane Street enforcement action suggests that regulatory interventions targeting institutional misconduct do not necessarily deter retail engagement. SEBI's recent research and regulatory initiatives are focused on limiting retail losses, increasing financial literacy, and deterministically analyzing the heterogeneity of retail participation. This could entail a breakdown of regional engagement, gender demographics, speculative versus hedging strategies (though the former dominates among retail players), and the impact of increasing experience levels. However, though these initiatives are progressing, in FY 2024-25, the net losses of retail traders widened to *Rs.*1.05 lakh crore (up 41% from *Rs.*74,812 crore in FY 2023-24).

Regarding effective mitigation, on one hand, aggressive enforcement against institutional manipulation may improve market integrity and reduce retail exploitation. On the other

hand, the disruption to institutional liquidity provision may create short-term market dislocations that retail traders attempt to exploit, potentially increasing their aggregate risk exposure even as they reduce specific position types. The combination of increased participation and altered risk-taking suggests that regulatory actions require careful calibration to avoid unintended consequences in retail-dominated markets. To substantiate and explore this implication, future studies can study the distribution of contract sizes among different participants and additional liquidity dynamics.

6 Conclusion

This study examines how retail traders in India's index options market responded to the July 2025 SEBI enforcement action against Jane Street, one of the world's largest quantitative trading firms. Using an event study methodology with 120-day pre-event baseline estimation, the analysis reveals that retail participation increased substantially and persistently (CAR of 26.89 percentage points by day 7), while retail put exposure decreased significantly (CAR of -13.41 percentage points by day 7). These findings challenge simple theoretical predictions and demonstrate that retail behavior in emerging market derivatives is more sophisticated and multidimensional than existing frameworks suggest.

The divergent patterns in participation and risk exposure reveal a retail population that, while frequently unprofitable, demonstrates awareness of changing market conditions and regulatory environments. The structural features producing 93% loss rates among retail options traders suggest that "strategic sophistication" does not translate into profitable outcomes.

More broadly, the Indian experience offers a laboratory for understanding retail behavior in markets where institutional dominance cannot be assumed. As electronic trading infrastructure, zero-commission brokers, and social media-driven financial literacy diffuse globally, other markets may increasingly exhibit India-like characteristics. The behavioral patterns

documented here may therefore represent not an exotic anomaly but rather a preview of retail dynamics in next-generation derivatives markets worldwide. Understanding how retail traders navigate institutional shocks in this environment provides essential groundwork for anticipating and regulating the evolving global derivatives landscape.

More broadly, the Indian experience offers a laboratory for understanding retail dominance in derivatives markets. As electronic trading infrastructure, zero-commission brokers, and social media-driven financial literacy diffuse globally, other markets may increasingly exhibit India-like characteristics with increasing retail influence.

Ultimately, stock markets emerged to provide companies with capital. With the advent of a vast array of financial products, the ecosystem of participating institutions broadened exponentially. In a world where wealth is accumulated among the few, the last 50 years has seen a rapid global democratization of innovation and ownership. Retail participation in derivatives markets continues to build upon this vision, providing broad access and risk management tools to all market participants while contributing to efficiency. Chief among these players are the ever-optimistic Indian retail traders, continuing to explore what the market has to offer amid the chaos of an ever-evolving global landscape.

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